



Oregon Institute of Technology

Computer Systems Engineering Technology Department

CST 407 – Python for AI

Homework 3

In this homework, you will be using the Speech Commands dataset. You can download the dataset from the following link:

https://oregontech-my.sharepoint.com/:u:/g/personal/fuat_karakaya_oit_edu/IQAYftM88rPnRr-Dla1rYpjAREKBC-_oWWGd6-sAdJhqSg?e=du2VdE

1. After downloading and uncompressing it, use **simple_audio.ipynb** notebook. Make sure DATASET_PATH points to the directory where the audio files are located.
2. Make sure every cell in the Jupyter notebook runs without any problem.
3. Now create your own WAV files for each command ('no', 'yes', 'down', 'go', 'left', 'up', 'right', 'stop').
4. Plot the waveforms of your WAV files over time and also plot their spectrograms.
5. Test your audio files with the trained model.
6. Now use keras-tuner to create a model. The following parameters must be tuned using keras-tuner:
 - a) Number of 2D convolution layers and number of filters for each convolution layer.
 - b) Number of Dense layers and number of Neurons for each Dense layer.

Model should also contain maxpooling layer(s) and Dropout layer(s).

7. Get the summary of the best model.
8. Test the same WAV files you have created using the best model obtained using keras-tuner.
9. Construct your first NN model: Use five conv layers and one dense layer before
10. Comment on the results
11. Submit your report, Jupyter notebook, and WAV files you have created.